MAT445

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(a) $\mathfrak{g}$ s.t. $\mathfrak{z}(\mathfrak{g}) = 0 \Rightarrow \exists\, \mathfrak{g} \hookrightarrow \mathfrak{gl}_n$

$$\mathfrak{z}(\mathfrak{g}) = \{x \in \mathfrak{g} : [x,y] = 0 \ \forall y \in \mathfrak{g}\}$$

$$\mathrm{ad} : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g})$$

$$x \longmapsto \begin{pmatrix} \mathrm{ad}_x : \mathfrak{g} \to \mathfrak{g} \\ y \mapsto [x,y] \end{pmatrix}$$

Lie hom as

$$\mathrm{ad}_x \mathrm{ad}_y - \mathrm{ad}_y \mathrm{ad}_x$$

$$\mathrm{ad}([x,y])(z) = [[x,y],z]$$
$$= -[[z,x],y] - [[y,z],x]$$
$$= [y,[z,x]] + [x,[y,z]]$$
$$= \mathrm{ad}_y([z,x]) + \mathrm{ad}_x([y,z])$$
$$= \mathrm{ad}_x \mathrm{ad}_y(z) - \mathrm{ad}_y \mathrm{ad}_x(z).$$

(b) $\mathfrak{g}$ abelian $\Rightarrow \exists\, \mathfrak{g} \hookrightarrow \mathfrak{gl}_n$.

$[x,y] = 0 \ \forall x,y \in \mathfrak{g}$.

(c) Cor to PBW. $\mathfrak{n}$ nilpotent $\hookrightarrow \mathfrak{gl}_n$.

Hint: $\mathfrak{n} \subset U\mathfrak{n}$, show $\exists\, m$ s.t. $\mathfrak{n}$ acts faithfully on $U\mathfrak{n}/I^m$.

$I = \langle \mathfrak{n} \rangle$

$$U\mathfrak{n} = \bigoplus_{i=0}^{\infty} \mathfrak{n}^{\otimes i} / \langle xy - yx - [x,y] \rangle \quad \text{assoc. k-alg}$$

$x_1 \cdots x_n$ basis $\mathfrak{n} \xrightarrow{\text{as k.v.s.}} x_1^{e_1} \cdots x_n^{e_n}$ basis $U\mathfrak{n}$ as k-v.s.

$$[\mathfrak{n}_p, \mathfrak{n}_q] \subset \mathfrak{n}_{p+q+1}$$
$$\| $$
$$[x,y] : \quad x = [$$

$\mathfrak{n}$ nilp.

$$\mathfrak{n} = \mathfrak{n}_0 \supset \mathfrak{n}_1 \supset \mathfrak{n}_2 \supset \cdots \supset \mathfrak{n}_{k-1} \supset \mathfrak{n}_k = 0. \qquad \mathfrak{n}_i = [\mathfrak{n}, \mathfrak{n}_{i-1}].$$

$\mathfrak{n}_{k-1}$ has basis $x_1 \cdots x_{n_1}$

$\mathfrak{n}_{k-2}$ $\quad x_1 \cdots x_{n_1} \cdots x_{n_2}$

$$v(x_1^{\alpha_i} x_j^{\beta_j}) = \alpha_i v(x_i) + \beta_j v(x_j) \qquad i \le j$$

$$i > j : \quad x_i^{\alpha_i} x_j^{\beta_j} = x_i^{\alpha_i - 1}(x_j x_i + [x_i, x_j]) x_j^{\beta_j - 1}$$

$\beta_j = 1$ ✓

$$d^{n+1} \circ d^n f(x_1 \wedge \cdots \wedge x_{n+2})$$

$$\beta^{n+1} \circ \alpha^n f(x_1 \wedge \cdots \wedge x_{n+2})$$

$$= \sum_{i<j} (-1)^{i+j} \alpha^n f([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_{n+2})$$

$$= \sum_{i<j} (-1)^{i+j} [x_i, x_j] f(x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_{n+2})$$

$$+ \sum_{i<j} (-1)^{i+j} \Bigg(\sum_{k=1}^{i-1} (-1)^k x_k f([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_k \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_{n+2})$$

$$+ \sum_{k=i+1}^{j-1} (-1)^{k-1} x_k f([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_k \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_{n+2})$$

$$+ \sum_{k=j+1}^{n+2} (-1)^k x_k f([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \cdots \wedge \hat{x}_k \cdots x_{n+2})$$

$$\alpha^{n+1} \circ \beta^n = \sum_{i=1}^{n+2} (-1)^{i+1} x_i \beta^n f(x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge x_{n+2}) = \sum_{i=1}^{n+2} (-1)^{i+1} x_i \Bigg[\sum_{j<k<i} (-1)^{j+k} f([x_j, x_k] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge x_{n+2})$$

$$+ \sum_{j<i<k} (-1)^{j+k+1} f([x_j, x_k] \wedge x_1 \cdots \wedge \hat{x}_i \wedge \cdots x_{n+2})$$

$$+ \quad (-1)^{j+k}$$

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$$\beta^{n+1} \circ \beta^n f(x_1 \wedge \cdots \wedge x_{n+2}) = \sum_{1 \le i < j \le n+2} (-1)^{i+j} \beta^n f([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_{n+2})$$

$$= \sum_{i<j} (-1)^{i+j} \Bigg[\underbrace{\sum_{k=1}^{i-1} (-1)^k f([[x_i, x_j], x_k] \wedge \cdots) + \sum_{k=i+1}^{j-1} (-1)^{k+1}}_{\text{Jacobi}} \Bigg]$$

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k char $p>0$. $\mathfrak{pgl}_p(k) := \mathfrak{gl}_p(k) / k \cdot I$ semisimple but not a product of simple Lie algebras.

$\hookrightarrow \underset{\text{max. solvable ideal.}}{\underset{\shortparallel}{\sqrt{\mathfrak{g}}}} = 0$

$\hookrightarrow \underset{[D^{N-1}\mathfrak{g}, D^{N-1}\mathfrak{g}]}{\underset{\shortparallel}{D^N \mathfrak{g}}} = 0$.

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Derived series $D^0\mathfrak{g} = \mathfrak{g}$, $D^{i+1}\mathfrak{g} = [D^i\mathfrak{g}, D^i\mathfrak{g}]$

- $\mathfrak{g}$ solvable if $D^N\mathfrak{g} = 0$.

Lower central series $C^0\mathfrak{g}$, $C^{i+1}\mathfrak{g} = [\mathfrak{g}, C^i\mathfrak{g}]$

- $\mathfrak{g}$ nilpotent if $C^N\mathfrak{g} = 0$.
- $D^i\mathfrak{g} \subset C^i\mathfrak{g}$ $\rightsquigarrow$ nilpotent $\Rightarrow$ solvable.

not converse: ex. $\mathfrak{b}_n = \{\text{upper } \nabla \text{ matrices}\}$ solvable but not nilpotent.
$\mathfrak{n}_n = \{\text{strictly upper } \nabla \text{ matrices}\}$ nilpotent.
$\Big\}$ since $[\mathfrak{b}_n, \mathfrak{b}_n] \subset \mathfrak{n}_n$, it will follow that $\mathfrak{b}_n$ is solvable.

$\sqrt{\mathfrak{g}}$ maximal solvable ideal

why does it exist? $I_1, I_2 \subset \mathfrak{g}$ solvable. $I_1 + I_2$ is idl, claim: solvable. note $D^i(\mathfrak{g}/\mathfrak{h}) = D^i\mathfrak{g}/\mathfrak{h} \cap D^i\mathfrak{g}$, so $\mathfrak{g}$ solvable $\Rightarrow$ $\mathfrak{g}/\mathfrak{h}$ solvable.

now $I_1 + I_2/I_1 \cong \underbrace{I_2/I_1 \cap I_2}_{\text{solvable}}$. ets if $\mathfrak{g}/I$ solvable + I solvable then $\mathfrak{g}$ solv.

indeed $0 = D^i(\mathfrak{g}/I) = D^i\mathfrak{g}/I \cap D^i\mathfrak{g} \to D^i\mathfrak{g} \subset I$
$\underset{D^iI = 0}{+} \Longrightarrow D^{i+j}\mathfrak{g} = 0$. $\square$

$\mathfrak{g}$ semisimple if $\sqrt{\mathfrak{g}} = 0$.
reductive if $\sqrt{\mathfrak{g}} = \mathfrak{z}(\mathfrak{g})$.

Thm (Lie) $k = \bar{k}$ char 0. $\mathfrak{g}$ solvable. Then $\mathfrak{g}$-reps are upper triangularizable.

- equivalently, $\exists$ flag $F = (0 = V_0 \subset V_1 \subset \cdots \subset V_n = V)$ st. $\rho(\mathfrak{g}) \subset \mathfrak{b}(F)$.
- by induction, if $V \neq 0$ $\mathfrak{g}$-rep then $\exists\, 0 \neq v \in V$ simult eigenvector for $\mathfrak{g}$.

Lem. $k = \bar{k}$ char 0. $\mathfrak{h} \subset \mathfrak{g}$. $V \in \mathfrak{g}$-mod. $0 \neq v \in V$. $\chi : \mathfrak{h} \to k$ st. $hv = \chi(h)v$ $\forall h \in \mathfrak{h}$ $\Rightarrow$ $\chi([x,h]) = 0$ $\forall x \in \mathfrak{g}, h \in \mathfrak{h}$.

Pf. $x \in \mathfrak{g}$. $V_i := \operatorname{span}_k\{v, xv, x^2v, \dots x^{i-1}v\}$.

$0 = V_0 \subset V_1 \subset \cdots \subset V_i$ let n min. st. $V_n = V_{n+1}$. then $xV_n \subset V_n$ and $V_n = V_{n+k}$ $k \geq 1$.

<u>claim</u>: $h \in \mathfrak{h} \rightsquigarrow hx^iv \equiv \chi(h)x^iv \mod V_i$ $\forall i \geq 0$.

Proof. by induction on i: $i = 0$: $hv = \chi(h)v$ ✓

$i > 0$: $hx^iv = hxx^{i-1}v = xhx^{i-1}v - [x,h]x^{i-1}v$

$= \chi(h)x^{i-1}v + v' - \underbrace{\underbrace{[x,h]}_{\in \mathfrak{h}, \text{ so}} x^{i-1}v}_{\in V_{i-1}}$, $v' \in V_{i-1}$.

$\equiv \chi(h)x^{i-1}v \mod V_i$. ✓claim.

now $\{v, xv, \dots, x^{n-1}v\}$ basis V_n. $h \in \mathfrak{h}$ acts triangularly on V_n,

$\operatorname{tr}(h|_{V_n}) = n\chi(h)$. so

$0 = \operatorname{tr}([x,h]|_{V_n}) = n\chi([x,h])$. $\square$

Pf (Lie) ind. on $\dim$. $\dim \mathfrak{g} = 0$ ✓

> 0: $D\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$.

$\subset \mathfrak{h} \subset$ codim 1.

$\mathfrak{h}$ is idl as $[\mathfrak{g}, \mathfrak{h}] \subset [\mathfrak{g}, \mathfrak{g}] \subset \mathfrak{h}$.

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Semisimple Lie Gps/Alg

all Lie algebras/$\mathbb{C}$.

1.1. $sl_2(\mathbb{C})$. $e = \begin{pmatrix} & 1 \\ & \end{pmatrix}$, $f = \begin{pmatrix} & \\ 1 & \end{pmatrix}$, $h = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$. $[h,e] = 2e$, $[h,f] = -2f$, $[e,f] = h$.

$\int$

$gl_2(\mathbb{C}) = gl(V)$. We know the irreps of $sl_2(\mathbb{C})$ are exactly $\mathrm{Sym}^N(V)$ $N \geq 0$.

Def. $\mathfrak{t} \subset L$ subalg is <u>toral</u> if consists of semisimple elements.

Lem. toral $\Rightarrow$ abelian

Proof. if not, $\exists\, x, y \in \mathfrak{t}$ st. $[x,y] = ay$, $a \neq 0$.

y acts on $\langle x,y \rangle$ by $\begin{pmatrix} & a \\ & \end{pmatrix}$ nilpotent, contradicting y ssimple $\square$

- $\mathfrak{t}$ Cartan (i.e. max toral subalg) of $\mathfrak{g}$
- decompose $\mathfrak{g}$ into $\mathfrak{t}$-eigenspaces $\mathfrak{g}_\alpha = \{ x \in \mathfrak{g} : [h,x] = \alpha(h)x \ \ \forall h \in \mathfrak{t} \}$.
  $\alpha \in \mathfrak{t}^\vee$

  i.e. x is eigenvect for $\mathfrak{t} \circlearrowright \mathfrak{g}$
  w/ eigenval $\alpha : \mathfrak{t} \to \mathbb{C}$.

$\Phi = \mathfrak{t}^\vee - 0$ roots

$\mathfrak{g}_\alpha$ weight/root space

$x \in \mathfrak{g}_\alpha$ root vectors

Prop.
(1) $\mathfrak{t} \neq 0$
(2) $K|_{\mathfrak{t}}$ nondegen. if $\alpha + \beta \neq 0$ then $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$. $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$.
(3) $\mathfrak{t} = \mathfrak{g}_0$
(4) $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$
↳ by (2), $\exists!\, t_\alpha \in \mathfrak{t}$ w/ $K(h, t_\alpha) = \alpha(h)$ $\forall h \in \mathfrak{t}$.
(5) $a \in \mathfrak{g}_\alpha$, $b \in \mathfrak{g}_{-\alpha}$ $\Rightarrow [a,b] = K(a,b)\, t_\alpha$
(6) $K(t_\alpha, t_\alpha) = \alpha(t_\alpha) \neq 0$.
(7) $\exists\, e_\alpha \in \mathfrak{g}_\alpha$, $f_\alpha \in \mathfrak{g}_{-\alpha}$, $h_\alpha \in \mathfrak{t}$ forming sl_2.

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